

Clinical Practice Guideline: Management of Type 2 Diabetes Mellitus

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Screening and Diagnosis

Adults aged 35 years or older without symptoms of hyperglycemia should be screened for type 2 diabetes at least every 3 years using fasting plasma glucose, HbA1c, or a 75-gram oral glucose tolerance test.

A diagnosis of diabetes is confirmed by an HbA1c of 6.5% or greater, a fasting plasma glucose of 126 mg/dL or greater, or a 2-hour plasma glucose of 200 mg/dL or greater during an oral glucose tolerance test.

Individuals with a body mass index of 25 kg/m² or higher who have one additional risk factor, such as a first-degree relative with diabetes or a history of gestational diabetes, should be screened at any age.

Pharmacologic Therapy

Metformin remains the preferred initial pharmacologic agent for most patients with type 2 diabetes, started at diagnosis alongside lifestyle intervention unless contraindicated by an estimated glomerular filtration rate below 30 mL/min/1.73m².

For patients with established atherosclerotic cardiovascular disease, a glucagon-like peptide-1 receptor agonist or a sodium-glucose cotransporter-2 inhibitor with demonstrated cardiovascular benefit should be added independent of HbA1c or metformin use.

The general HbA1c target for most nonpregnant adults with diabetes is below 7.0%, though a less stringent target of below 8.0% may be appropriate for patients with limited life expectancy or extensive comorbidity.

Monitoring and Complications

Comprehensive foot examinations, including assessment of pulses, sensation, and skin integrity, should be performed at least annually for all patients with diabetes.

Urine albumin-to-creatinine ratio and estimated glomerular filtration rate should be assessed at least once per year in all patients with type 2 diabetes to screen for diabetic kidney disease.

A dilated eye examination for diabetic retinopathy should occur within five years of diagnosis for type 1 diabetes and at the time of diagnosis for type 2 diabetes, then at least every two years thereafter if no retinopathy is found.

Special Populations

In pregnant patients with pre-existing diabetes, the target HbA1c prior to conception is below 6.5% when it can be achieved without significant hypoglycemia.

Screening for gestational diabetes should be performed at 24 to 28 weeks of gestation using either a one-step 75-gram oral glucose tolerance test or a two-step approach with a 50-gram screen followed by a 100-gram confirmatory test.

In older adults with diabetes and significant comorbidity or cognitive impairment, an HbA1c target of below 8.0% to 8.5% is reasonable to reduce the risk of hypoglycemia.

Lifestyle Management

Patients with type 2 diabetes should aim for at least 150 minutes per week of moderate-to-vigorous aerobic physical activity, spread over at least 3 days per week with no more than 2 consecutive days without activity.

Medical nutrition therapy delivered by a registered dietitian should be offered to all patients with diabetes at diagnosis and reassessed at least annually.

Blood Pressure and Lipid Management

Blood pressure should be measured at every routine clinical visit, with a general target of below 130/80 mmHg for most patients with diabetes and hypertension.

A moderate-intensity or high-intensity statin should be prescribed for all patients with diabetes aged 40 to 75 years, in addition to lifestyle therapy, regardless of baseline lipid levels.